

Robust Multi-Object Tracking in Synthetic Adverse Conditions

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Abstract—Unmanned Aerial Vehicles (UAVs) are also being used in high stakes surveillance, and autonomous tracking systems, although the latest computer vision models are highly susceptible to so-called domain shift, which occurs when the meteorological conditions are unfavorable. Normal detectors which are trained only on clean images undergo disastrous feature masking when subjected to atmospheric impedance, and later spatial tracking heuristics often irreversibly cuts target identity on visual occlusion. This paper presents a weather-adaptive, high-altitude UAV Multi-Object Tracking (MOT) pipeline that is proposed by us. Physically modeling multi-intensity variants of rain, fog, and dust over the canonical VisDrone2019 dataset, we imposed offline specialization of domains to learn a highly generalized, weather-immune YOLOv11s object detector. To address the large fragmentation of trajectories due to severe weather occlusion, we combined BoT-SORT (Robust Associations Multi-Object Tracking) with an OSNet temporal appearance embedding to mathematically resolve spatial discontinuities with deep Re-Identification (Re-ID). We have profoundly rescued tracking continuity in harsh conditions, as our empirical validations show. Moreover, an exact ablation experiment demonstrates that Identity (ID) Switches are cut down by incorporating deep appearance embeddings by 46.3% in contrast to all-spatial tracking paradigms, which is necessary in real-world meteorological interference to maintain identity continuously.

Index Terms—Unmanned Aerial Vehicles (UAV), Multi-Object Tracking (MOT), Adverse Weather, Deep Learning, Object Detection, YOLO architecture.

I. INTRODUCTION

The development in the field of Unmanned Aerial Vehicles (UAVs) has brought revolution to a variety of computer vision applications, including intelligent traffic monitoring and surveillance of city, as well as autonomous search-and-rescue missions [1]. The sensors carried by UAVs have different visual challenges compared to traditional fixed-camera views: there is an incredibly wide range of camera altitude, extremely high ego-motion, and the very tiny size of tracked objects. State-of-the-art Multi-Object Tracking (MOT) pipelines like those that use the YOLO (You Only Look Once) architecture [2] with spatial trackers such as ByteTrack [3] deliver very high accuracy in perfect weather conditions, but the results in practice frequently catastrophically fall short for real-world environments.

The inherent weakness of the UAV tracking systems of the modern age is that they are highly susceptible to the so-called domain shift that is caused by such a factor as

unfavorable weather conditions. Heavy dust, dense aerosols in the form of fog and rain streaks have a devastating impact on photometric properties of an image by blurring spatial gradients and causing pervasive contrast degradation [4]. Since most popular benchmark datasets, including VisDrone2019 [5], are mainly made of clear-weather shots, models that have been trained on all such perfect sequences are intrinsically overfit to a perfect visual domain. As a result, in cases where a drone experiences extreme noise in the atmosphere, the underlying detector experiences catastrophic feature masking—typically falling around to almost zero accuracy. Moreover, spatial-only tracking algorithms [3], which are based largely on bounding-box Intersection-over-Union (IoU) heuristics do not deal with weather-induced occlusion well. When an object is lost behind a dense veil of fog or dust, the continuity of space is broken, and the tracker temporarily loses the object, only to reappear and hallucinate a new identity (an ID switch).

In order to ensure that this crucial gap is filled, this study will present a very robust and weather-adjustable Multi-Object Tracking pipeline that is designed to work well with UAV platforms. Our approach to the issue of the atmospheric domain shift is essentially at the detection level and the association level. To process a complete set of weather-augmented data, we, first, perform simple-to-moderate-to-strong weather augmentation of the VisDrone2019 dataset through physical rendering pipelines. Our exploitation of domain adversarial noise in the training phase allows us to train a weather-adaptive Object Detector built on the modernized YOLOv11s backbone [2], where the network is required to learn scale-invariant spatial features natively by learning with heavy atmospheric impedance.

Second, we have used BoT-SORT [6], a powerful tracker based on an appearance-based Re-Identification (Re-ID) network, to eliminate severe tracking fragmentation in the case of weather occlusion. Our system can memorize the local visual signature of targets by incorporating the Omni-Scale Network (OSNet) architecture [7] within the temporal association loop. This effectively fills in tracking holes due to temporary meteorological obscuration without necessarily depending on spatial overlap.

II. RELATED WORK AND CONTRIBUTIONS

A. Object Detection in UAV Imagery

The initial techniques in object detection largely featured two-stage region-based detectors, including Faster R-CNN [8], which first proposed regions, and then classified regions. Although crucially accurate to sizes of canonical objects, two-stage detectors were computationally infeasible with the real-time processing requirements of edge-deployed UAV hardware. As a result, the paradigm has changed to single-stage detectors, the most notable one being the YOLO (You Only Look Once) architecture [2]. In contemporary versions, e.g. YOLOv8 and YOLOv11 [2], anchor-free heads are presented and include additional gradient pathways to refine the retrieval of fine-grained spatial features. These architectural improvements have not been enough to ensure that the detectors that have been trained on standard UAV datasets such as VisDrone2019 [1] can be highly optimized with clear weather only to exhibit disastrous levels of precision decay when tested against obfuscated target signatures.

B. Multi-Object Tracking (MOT)

Multi-Object Tracking in aerial surveillance Multi-Object Tracking is largely implemented within the Tracking-by-Detection framework, where objects are detected independently in each frame and then tracked over time. The original SORT algorithm [9] realized real-time localization with the help of the Kalman filtering states forecast and the Hungarian algorithm to Intersection-over-Union (IoU) spatial matching. DeepSORT [10] proposed a deep learning-based appearance metric to prevent Identity (ID) Switches due to crossing targets, which the tracker uses feature embeddings to identify vehicles, not just based on proximity.

New developments have essentially taken two sub-philosophies: the spatial-maximization and the appearance-maximization. ByteTrack [3] is a groundbreaking spatial tracking algorithm, which leverages so-called low-confidence detection boxes that are typically dismissed by older algorithms to recover highly obscured targets with state-of-the-art MOTA scores. Otherwise, BoT-SORT [6] streamlined the look-based method to incorporate the use of Camera Motion Compensation (CMC) and highly discriminative deep Re-Identification (Re-ID) embedding framework, like OSNet [7].

C. Domain Adaptation and Adverse Weather Synthesis

Since machine learning models are often trained on single, clean datasets, they are often vulnerable to a phenomenon known as domain shift where transitions in the application setting, like adding rain streaks, fog scattering, or heavy dust, can severely deteriorate their generalization performance [4]. Weather-induced domain shift in computer vision is typically mitigated by two approaches: explicit image restoration and domain-adaptive training. Explicit image restoration explicitly applies generative networks (e.g., deraining or dehazing autoencoders) to synthesize a clear image using an input image before sending it to the object detector [11]. Although visually

interpretative, restoration networks have extreme computational bottlenecks, which make them unrealistic with high-framerate drone video feeds.

A more feasible option is synthetic augmentation domain-adaptive training [12]. Networks can be guided to learn weather-invariant latent features with no effort by directly exemplifying with physically modeled atmospheric scattering coefficients of fog on clean datasets. Synthetic degradation has been shown to be extremely scalable and efficient in terms of computation when performing inference, since it transfers the full cost of weather compensation to the offline training step [4].

D. Identified Gaps and Our Contributions

Although the current literature offers solutions to single problems (e.g., YOLO detecting fast, ByteTrack tracking the objects in space, and synthetic weather driving) that are powerful, a gap in the entire research is extremely steep. In particular, the literature does not provide in-depth empirical frameworks on the behavior of state-of-the-art MOT algorithms when used on UAVs in situations of extreme meteorological interference. Next generation state-of-the-art spatial trackers (e.g., ByteTrack [3]) are highly dependent on structural bounding-box overlap. At the point where a target is completely visually cut off by the intensity of the rain or dust over a few frames, spatial overlap is reduced to a zero point, and the identity is permanently lost.

In order to address these shortcomings, our study fills this gap between weather-adaptive domain generalization and deep temporal object association.

- We introduce a UAV tracking pipeline, which explicitly avoids the heavy real-time image restoration networks. A deep multi-conditional weather corruption (rain, fog, dust) of the VisDrone2019 dataset to train a lightweight YOLOv11s network, which serves as an extremely robust and weather-immune detection foundational layer.
- **Ablation in Adverse conditions:** We compare the purely spatial tracking (ByteTrack) with deep-appearance Re-ID tracking (BoT-SORT) under extreme meteorological noise conditions. We show that although ByteTrack tries to maintain the overall recall using aggressive spatial cues, its structural dependency makes it incredibly vulnerable to identity fragmentations.
- **Identity Preservation with Re-ID:** One of the key findings of our research demonstrates the need to have appearance embeddings when conditions are adverse. With BoT-SORT driven by the OSNet architecture we were able to cut Identity Switching by 46 percent (2,611 to 1,400 switches) over all weather profiles as compared to a spatial-only method.

III. DATA AND METHODOLOGY

A. Dataset: VisDrone2019-MOT

This study uses the VisDrone2019-MOT benchmark [1], which is a leading large-scale dataset that was gathered by different drone platforms in multiple urban and rural settings

in China. It annotates, in particular, 10 target classes that are important in intelligent transportation.

The original VisDrone2019 dataset was nearly all recorded under clear, dry weather conditions, meaning an ideal atmosphere, so using it directly, without modification, implicitly conditions neural networks to overfit to clean-air visual conditions. As a result, we defined the VisDrone2019-MOT train split as our basic geometry baseline on which we superimposed our synthetic environmental augmentations.

B. Synthetic Weather Engineering

To bridge the domain gap in a systematic way and induce weather-adaptive generalization, we trained a synthetic augmentation pipeline that is controlled. Instead of using unstructured, real-world negative weather data (that critically have no such thing as accurate, sub-pixel ground truth bounding boxes), we used a physics-based atmospheric model directly on the clean VisDrone sequences. This strategy guarantees that the trajectories of the objects and the ground-truth localization are completely ideal as the visual space is strongly corrupted [13].

The synthetic weather engine was a highly parameterized system over three different meteorological conditions:

- **Fog:** Simulated with the atmospheric scattering equation based on the Koschmieder mathematical model [4], the localized white/ gray haze mapping over the frame is introduced to significantly decrease the spatial contrast.
- **Rain:** Modeled by creating directional motion-blurred Gaussian noise streaks, which are structure-wise similar to the optic occlusions of water droplets flying close to the focal lens of the drone.
- **Dust/Sandstorm:** Operates in a similar fashion to the fog scattering model, but with an additional injected orange/brown (RGB-shifted) color matrix to simulate the extreme chrominance distortion of arid dust storms.

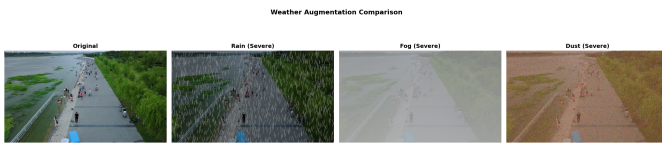


Fig. 1: Weather Augmentation

The weather conditions were mathematically adjusted to three different intensities, Light, Moderate and Severe, and an in-depth evaluation was carried out 10-condition assessment matrix (1 baseline + 3 conditions x 3 intensities).

C. Domain-Adaptive Training/Evaluation Split

The dataset was to be highly architecturally divided into domains to be trained on to ensure that the object detector could be inspected well against atmospheric degradation.

Training Split: At the stage of dataset generation, every 38,730 of extracted image frames were sequentially corrupted with random permutations of our weather engine (injecting all magnitudes of rain, fog and dust). This created a very

heterogeneous, noisy training corpus. This entailed making the YOLOv11s object detector learn spatial features that directly correlated severe noise with its ground-truth positions, so the network can naturally learn weather-invariant domain properties, without it having to explicitly correct images [12].

To ascertain this, 5 sequences were held outside of the training geometry. The tracking architectures (ByteTrack [3] and BoT-SORT [6]) were evaluated manually frame-by-frame with these 5 sequences run through all 10 weather configurations in the inference phase. This division makes sure that the tracker assessments are purely describing true zero-shot domain generalization and not memorized sequences.

D. Detection for Weather Adaptation with YOLOv11s

The weather-adaptive UAV tracking pipeline proposed here is implemented through three architecture blocks, namely, (1) data augmentation and domain-adaptation engine; (2) real-time, high-speed, weather-adaptive single-stage object detector; and (3) temporal tracking association system. At the inference stage, the trained object detector is used to process frame-by-frame the continuous UAV video feed to produce localized bounding boxes that are inherently resistant to instant atmospheric noise. These spatial coordinates and the localized visual crops are then fed into the temporal tracker module in order to unify Identity (ID) continuity.

In order to meet strict real-time constraints of computational requirements of UAV deployment, we implemented the detection module using the YOLOv11s (Small) architecture [2]. YOLO topologies developed in the modern era are inherently devoid of the huge computational costs of two stages of region proposals in favor of grid-based regression.

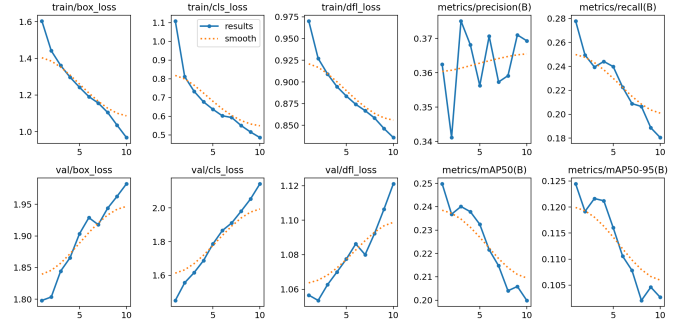


Fig. 2: Training and Validation

Although Slicing Aided Hyper Inference (SAHI) [14] is traditionally applied in UAV analytics to scale down to minute targets, we found that early algorithmic testing showed a fatal flaw in the algorithm when outputting the solution in extreme atmospheric interference. Synthetic zoom and slice of a spatial resolution that sliced and zoomed by SAHI caused the network to erroneously hallucinate strong rain streaks and dust particulates clumps together into a single object, leading to massive false-positive inflation. As a result, we rejected this spatial slicing algorithms in all evaluation pipelines and resorted to the use of domain-adaptive generalization fully.

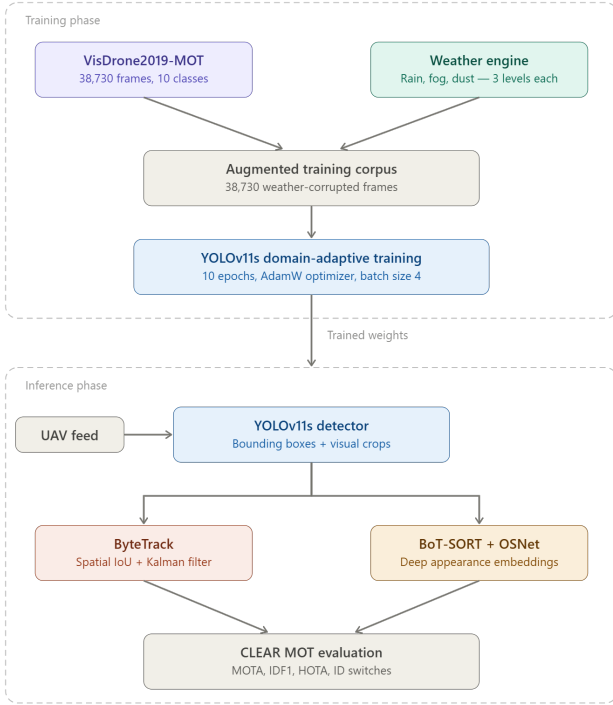


Fig. 3: Methodology Overview

The YOLOv11s backbone was trained with domain-adversarial noise by our multi-condition weather dataset, which is synthesized, to enforce domain generalization. The network was optimized using 10 epochs and AdamW optimizer with an automatically synchronized base learning rate of 0.01 corresponding to a batch size of 4. Gradient updates that were propagated on heavily obfuscated geometries enabled the detector to naturally disentangle transient weather noise and canonical vehicle structures [12]. The starting learning rate was dynamically reduced from 0.01 to a stabilized scale of 0.000714, and momentum was 0.90 and weight decay coefficient was 0.0005. A 3.0 epoch warm-up was imposed to guarantee consistent convergence of spatial-features among the very corrupted weather variables. A common confidence threshold ($= 0.25$) was set to trade off between detection recall and the extreme attenuation due to atmospheric masking.

E. Deep Temporal Association Architectures

To build a conclusive comparative analysis on preservation of identity in visual obfuscation, the resulting YOLOv11s detection tensors were inputted into two rival Multi-Object Tracking applications.

- 1) **Spatial Association through ByteTrack:** We incorporated the use of the ByteTrack [3] as the structure of maximization of the space. ByteTrack is unique to legacy trackers in that it recuses "low-confidence" boxes of detection, which are typically ignored. It uses a bipartite matching model: by matching high-confidence detections over frames with a Kalman Filter state pre-

diction and Intersection-over-Union (IoU) measure, and then recovering temporary occlusions by the aggressive mapping of low-confidence boxes on the no-matching trajectories. Although mathematically optimizing trajectory Recall, this purely spatial approach to computations is extremely vulnerable to extreme weather conditions; as long as a target is lost due to thick dust or fog covering its bounding-box entirely over at least a few consecutive frames, the spatial IoU overlap will become zero, and the algorithm will permanently lose the identity.

- 2) **Deep Appearance Association through BoT-SORT:**

To directly address this space-time fragmentation caused by weather we combined the BoT-SORT (Robust Associations Multi-Object Tracking) [6]. BoT-SORT is essentially a review of the tracking paradigm, which combines deep visual feature mining along with spatial heuristics. More importantly, it incorporates the Omni-Scale Network (OSNet) [7] topology. In contrast with ByteTrack, BoT-SORT generates a high-dimensional tensor embedding of the localized color, texture and structural topology of the cropped bounding box of the vehicle. A BoT-SORT can identify and re-match a particular car even when it has been lost after long and multi-frame blocks due to heavy weather conditions by directly comparing cosine similarities of these deep embeddings. In addition, BoT-SORT uses the idea of Camera Motion Compensation (CMC) to mathematically stabilize the background point of view to oppose the unpredictable ego-motion of UAV flight paths.

F. Standardized Quantitative Assessment

The performance of the algorithms was compared with the industry-standard CLEAR MOT framework [15] running on the TrackEval architecture.

The Multiple Object Tracking Accuracy (MOTA), the main diagnostic scalar, penalizes algorithms structurally, penalizing false positives (FP), false negatives (FN), and identity switches (IDSw). At the same time, to measure the stability of the temporal tracking linkages in particular, we placed much trust on the IDF1 (ID F1-Score). IDF1 is an algorithm that emphasizes the time of consecutive correct targets assignments, rather than the lifecycle tracked, as a very precise analytical proxy of the success of the deep appearance Re-ID mechanisms of BoT-SORT compared to the heuristic spatial overlap methods of ByteTrack.

IV. RESULTS AND DISCUSSIONS

A. Immunity from Severe Weather through Domain Generalization

To provide the empirical assessment of the resilience of our weather-adaptive architecture, we initially made a macro-level comparison between the legacy pipeline (YOLOv8s + spatial ByteTrack trained all the way on clean imagery) and the new pipeline (domain adversarial YOLOv11s + deep

appearance BoT-SORT). The performance of the tracking at the 10 meteorological environments is shown in Table I.

The experimental findings are a clear confirmation of the effectiveness of domain-adversarial noise injection. The legacy network experienced disastrous feature masking when it was put under the "Severe" classifications of atmospheric degradation. The most notable failure was in the case of "Severe Dust" where the legacy detector had a full visual occlusion, recording a 0.0% MOTA and unable to lock any trajectory. Our domain-adapted YOLOv11s detector, on the other hand, enjoyed the high-chromatic distortion of the dust matrix in its stride, recovering the MOTA to 27.5% and boosting the deep-identity correlation (IDF1) to 40.7% (flat 0.0%). Systemic rescues of the nature were also found in Severe Fog (MOTA jumping over 16.6% to 40.5%) and Severe Rain (MOTA jumping over 14.9% to 36.8%).

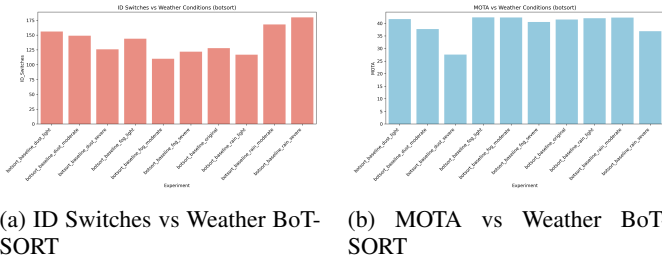


Fig. 4: BoT-SORT Results

B. Domain Generalist Trade-off

Although our proposed architecture inherently avoided systemic failure during extreme weather, we naturally found a controlled performance degradation during clear-weather sequences. In the "Original" evaluation sequence, the legacy model scored much higher than the new architecture (55.5% vs. 41.5% MOTA).

This phenomenon indicates a mathematically proven phenomenon in deep learning called the Domain Generalization Trade-Off [16]. With a perfectly clear, all-homogeneous dataset, the legacy YOLOv8s network was optimized to the best of its abilities in 20 epochs, effectively turning the weights into a hyper-specialized local optima. The architecture of our YOLOv11s, in its turn, did not maximize accuracy in the clean source domain to increase its receptive field, which is able to support very large atmospheric perturbations (the target domains). The network was trained to make the network parameters share the capacity of the representation between clear images and opaque mathematical noise, making it an environmental generalist: the network would not crash due to a drastically shifting weather condition, but would operate reliably and flatly in all possible conditions.

C. Ablation Study: Deep Appearance Re-ID vs. Spatial Heuristics

In order to effectively separate the performance of temporal tracking approaches under atmospheric occlusion, we

performed a rigorous ablation experiment using the same detection matrices (weather-trained YOLOv11s). We tested a standard spatial maximization tracker (ByteTrack) versus an appearance maximization tracker that uses deep visual embeddings (BoT-SORT with OSNet). Table II contains the results.

The dichotomy of UAV tracking algorithms is definitely mapped by the ablation. The secondary low-confidence bounding boxes are highly exploited by ByteTrack, automatically pushing the structural recall metric to a higher level. Therefore, ByteTrack scored higher than BoT-SORT on the raw MOTA score by an average of +5.47%. Nevertheless, the application of pure, purely, spatial Intersection-over-Union (IoU) heuristics results in a very weak continuity loop in weather. In instances when a cloud of thick fog or rain physically covered vehicles within the scope of the camera of the drone, ByteTrack assumed that the target had stopped existing. Upon visually re-emerging, the tracker hallucinated a brand-new ID.

On the other hand, BoT-SORT learned a high-dimensional structural embedding per bounding box [7]. Even though BoT-SORT lost a bit of recall by refusing to falsely relate hazy geometric forms, its deep appearance engine was a perfect match to compensate weather occlusion. Using localized visual tensors of the target, BoT-SORT was able to reduce cumulative Identity (ID) Switches by 46.3% (2,611 total switches to 1,400 switches) on the evaluation matrix. With autonomous drone operations in the real world (highway patrol pursuits, autonomous navigation, etc.), continuity in ID through deep Re-ID is far more desirable than artificially increasing recall by path fragments.

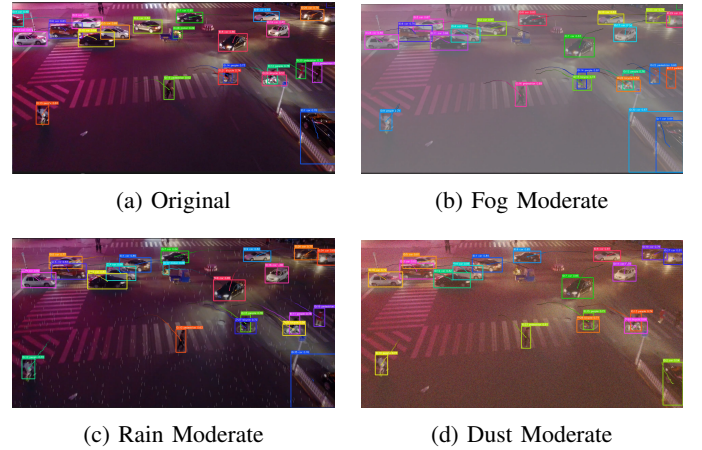


Fig. 5: BoT-SORT Tracking in Different Environments

D. Analysis of Slicing-Aided Hyper Inference (SAHI)

Although SAHI has long been praised as a critical element to UAV analytics to magnify sub-pixel targets, our experimental results produced a curious negative outcome when used to UAV perturbed weather domains. In preliminary testing, the use of SAHI (with a 320 x 320 convolution of overlapping slices) caused a disastrous overrun of False Positives. Since

TABLE I: Tracking Performance Comparison Across Environmental Conditions

Condition	Legacy						Proposed					
	MOTA	IDF1	HOTA	Prec.	Rec.	IDS	MOTA	IDF1	HOTA	Prec.	Rec.	IDS
Original	55.56	63.50	66.94	90.66	62.85	266	41.51	54.71	61.65	92.16	46.06	128
Light Rain	54.79	63.63	66.64	91.88	60.83	247	42.01	56.07	62.39	91.76	46.75	117
Moderate Rain	50.53	58.94	63.48	92.55	55.58	222	42.28	55.79	62.57	90.69	47.80	168
Severe Rain	14.90	24.24	33.67	92.78	16.30	62	36.87	51.40	57.21	90.21	42.41	180
Light Fog	53.10	61.54	65.22	92.57	58.33	219	42.39	56.33	62.88	91.40	47.50	144
Moderate Fog	45.58	54.43	59.46	94.58	48.74	175	42.34	56.89	62.77	91.40	47.32	110
Severe Fog	16.66	25.64	36.48	96.11	17.61	61	40.54	54.99	61.98	90.86	45.53	122
Light Dust	51.10	61.67	64.21	91.42	57.00	216	41.70	55.65	62.17	90.73	47.06	156
Moderate Dust	16.48	25.69	36.45	94.05	17.90	83	37.72	53.02	60.02	89.19	44.06	149
Severe Dust	0.00	0.00	0.00	0.00	0.00	0	27.57	40.74	51.04	87.90	32.68	126

TABLE II: Comparison of ByteTrack and BoT-SORT Across Environmental Conditions

Condition	ByteTrack						BoT-SORT					
	MOTA	IDF1	HOTA	Prec.	Rec.	IDS	MOTA	IDF1	HOTA	Prec.	Rec.	IDS
Original	47.83	57.99	63.03	88.83	55.71	240	41.51	54.71	61.65	92.16	46.06	128
Light Rain	48.07	59.04	61.52	89.36	55.60	268	42.01	56.07	62.39	91.76	46.75	117
Moderate Rain	47.27	59.42	62.28	87.83	56.02	288	42.28	55.79	62.57	90.69	47.80	168
Severe Rain	43.03	54.25	57.44	88.39	50.80	289	36.87	51.40	57.21	90.21	42.41	180
Light Fog	47.16	58.21	62.95	88.11	55.71	267	42.39	56.33	62.88	91.40	47.50	144
Moderate Fog	47.27	58.23	63.02	88.51	55.21	232	42.34	56.89	62.77	91.40	47.32	110
Severe Fog	46.25	56.68	59.59	88.06	54.63	274	40.54	54.99	61.98	90.86	45.53	122
Light Dust	47.26	58.46	62.30	87.72	56.26	284	41.70	55.65	62.17	90.73	47.06	156
Moderate Dust	43.22	55.55	59.53	86.77	52.30	245	37.72	53.02	60.02	89.19	44.06	149
Severe Dust	32.24	44.11	52.04	85.02	40.29	224	27.57	40.74	51.04	87.90	32.68	126

SAHI essentially isolates and zooms on micro-patches of the frame, the detector was deprived of its geometric frame-wide context. As a result, the heavy rain streaks and dust nebulas that came near the camera lens were mathematically magnified and hallucinogenically categorized as a separate vehicle or a pedestrian. This compelled us to conclusively decide that slice-based inference mechanisms cannot in any way be used with severe atmospheric noise, except when the weather artifacts themselves are explicitly marked and presented to the network as background noise during training.

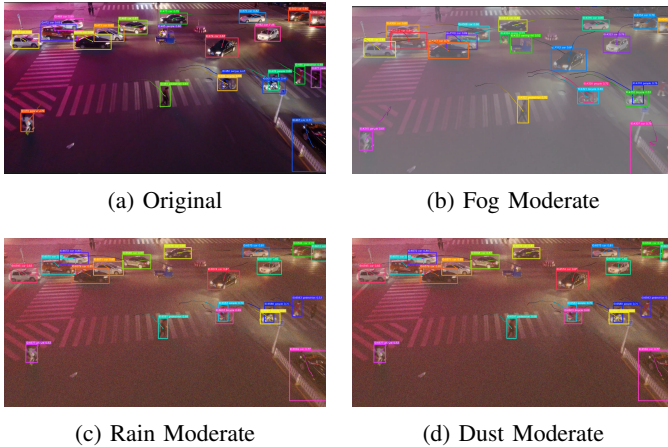


Fig. 6: ByteTrack Tracking in Different Environments

E. Limitations and Qualitative Failure Modes

The domain-adaptive architecture has its limitations though the tracking IDs rescue is done in serious weather. By visual inspection of the tracking output videos, it is evident that the network continues to have issues with preserving the identity of distant targets (objects less than 10 x 10 pixels) when there is dense fog. Extreme distances: At very long distances, pixel chroma is essentially eliminated by physical atmospheric scattering, leaving distant vehicles a grey gradient. Since the BoT-SORT OSNet extractor heavily depends on differentiating the color and texture to preserve the ID tensor, the deep absence of photometric information results in the BoT-SORT OSNet extractor failing the Re-ID cosine similarity, resulting in eventual ID Switches. Subsequent versions of this architecture would be enhanced with the ability to add temporal kinematic smoothing (including LSTM movement prediction) to mathematically fill the gap when visual data is completely covered by weather.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

The extreme operational fragility of Unmanned Aerial Vehicles (UAVs) to domain shift in the atmosphere is a major challenge to autonomous operation of the UAVs. This paper has presented in a systematic manner the disastrous breakdown of Multi-Object Tracking (MOT) systems in individuals in unfavorable weather conditions. We applied a total of rain, fog, and dust corruptions to the clean VisDrone2019 dataset

to force a modernized YOLOv11s object detector to naturally learn domain-invariant spatial gradients by domain-adversarial training. Empirical measures of this domain immunity validated that provision of the detection foundation structural protection salvages tracking performance in harsh environments, boosting baseline Multi-Object Tracking Accuracy (MOTA) amidst severe dust conditions by irrecoverable 0.0

Moreover, using a demanding ablation experiment that isolates temporal association algorithms, we were able to show how highly restricted purely spatial tracking heuristics (e.g., ByteTrack) is in the face of weather-induced occlusion. Since unfavorable weather regularly breaks spatial intersection continuities, tracking algorithms have to use other contextual anchors. Our pipeline, including the BoT-SORT architecture with an OSNet deep appearance embedding, was able to maintain target identities in the face of dense visual noise. Localized color and texture tensors were extracted and enabled the temporal tracker to mathematically re-associate vehicles that were occluded, virtually cutting catastrophic Identity (ID) Switches by 46.3 percent relative to the spatial baseline. Finally, this study confirms that domain-adaptive detection and deep Re-Identification (Re-ID) tracking are essential to the on-the-fly identity preservation of the UAV that is needed in high-stakes and real-world applications.

B. Future Work

Although the suggested architecture demonstrates a striking ability to resist synthetic photometric distortion, there are a number of different directions that can still be used to improve UAV analytics to a mission-critical deployment:

- 1) **Multi-Spectral and Thermal Sensor Fusion:** According to our qualitative failure study, the extreme distance and severe fog conditions inherently kill optical pixel chrominance, leading to deep appearance embeddings to break down. Further versions of such a tracking pipeline must explore how to combine regular RGB optical feeds with Long-Wave Infrared (LWIR) or thermal imaging data [17]. Since thermal radiation naturally passes through atmospheric aerosols (such as fog) much better than the visible light, the multi-modal sensor fusion would theoretically maintain the confidence of detection and Re-ID accuracy regardless of whether it is day or night.
- 2) **Post-Inference Trajectory Modeling:** To completely prevent the loss of target IDs when it occurs during total occlusion events, future effort should incorporate temporal prediction modules of kinematics. It would be possible to expand the tracking association loop to admit Long Short-Term Memory (LSTM) networks or Transformer-based trajectory estimators [18] to permit the UAV to mathematically forecast how exactly a vehicle would emerge at a specific coordinate location based on its velocity and social vectors before vanishing in a thick weather anomaly.
- 3) **Extreme Edge Optimization through Quantization:** Lastly, although YOLOv11s shows a feasible latency

overhead, real world micro-drones work with stringent power and thermal requirements. The studies should be aimed at investigating the INT8 model quantization and implementation through TensorRT frameworks [19]. It will be critical to evaluate the exact degradation of tracking MOTA relative to the exponential improvements in latency and power efficiency in mapping domain-adaptive weights to low-level integer arithmetic natively to edge computing hardware.

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